# MIKE3 Hydrodynamic model of Øresund

This repository contains practical and background information for the WaterBench case "MIKE3 HD model of Øresund".

The dataset can be cited as:

> DHI. (2026). Hydrodynamic 3D model of the Øresund: MIKE3 model setup, outputs and observation data [Data set]. Zenodo. https://doi.org/10.5281/zenodo.20284684

See the LICENSE file for usage conditions.

Disclaimer

> This model setup and dataset are provided for educational, benchmarking, and research purposes only.
> The model configuration is simplified and may not be fully calibrated or validated for operational use.
> Results must not be used for real-time forecasting, infrastructure design, safety-critical decisions, or regulatory purposes.
> The dataset does not represent best-practice modelling standards at DHI and should be treated as a controlled benchmark case.

## Intended Use

The dataset is aimed at supporting educational, research, and exploratory activities, such as:

- Development and benchmarking of hydrodynamic or hydrological models
- Testing model calibration and validation workflows
- Machine learning and surrogate modelling
- Extreme event forecasting studies
- Uncertainty quantification experiments

## Folder Structure

```
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- LICENSE
- model/
    + model setup files
    + simulation logs
- input/
    + boundary conditions
```

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    + meteorological forcing
    + diffusivity files
    + initial conditions
    + mesh file
- observations/
    + in-situ observations
    + satellite data
- code/
    + notebooks for data exploration
    + validation scripts
- figures/
- output_sample/
    + reduced model outputs for testing
- output/
    + full simulation results (only on Zenodo)
```

## Background

This folder contains data for a simulation with the hydrodynamic MIKE 3 Flow Model FM in the Øresund Strait between Denmark and Sweden. It is an extension of a 2D setup published in 2024 (https://doi.org/10.5281/zenodo.14160710). The repository contains the setup file, input data and observation data, as well as simulation result data. The simulation result data is available in Zenodo. The data supplied in this repository covers the period from 2023-01-01 to 2025-12-31.

### MIKE 3 Flow Model FM

MIKE 3 Flow Model FM is a 3D modelling system using a flexible mesh and a combination of a sigma and a Cartesian coordinate system. The modelling system has been developed for applications within oceanographic, coastal and estuarine environments. The governing equations are the incompressible Navier-Stokes equations subject to the assumptions of Boussinesq (density variations are small enough to be ignored, except in gravity term) and with the free surface described by a height function. The current model setup also includes the transport equations for temperature and salinity, which is useful e.g. for ecological modelling or modelling of sediment transport. More information can be found in the documentation.

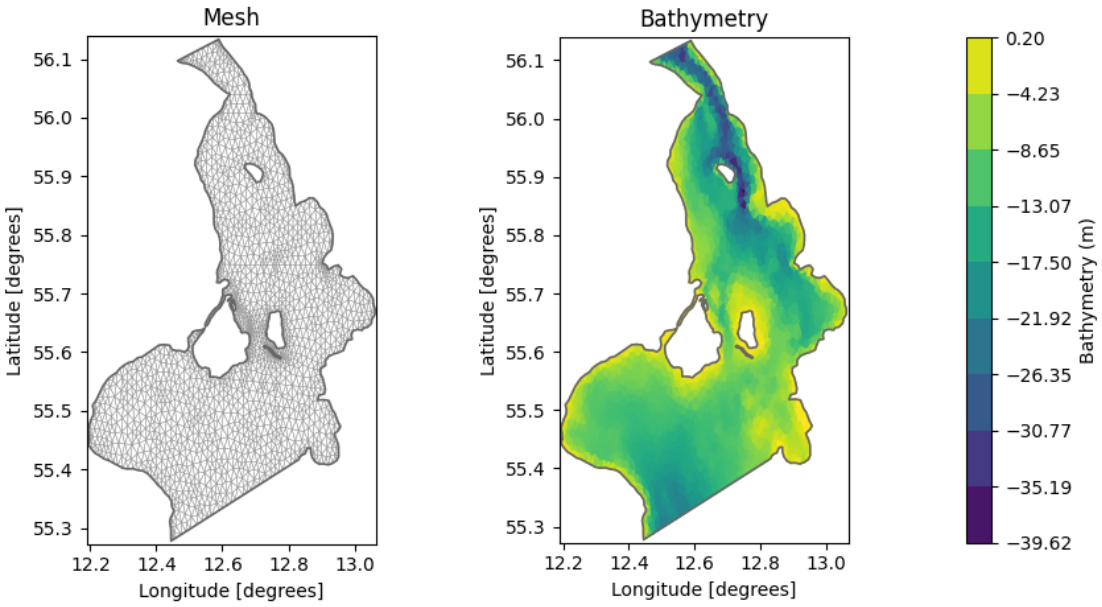
Since all data, including output from a simulation, is given, it is not necessary to run the MIKE simulation. However, it is possible, since a setup file is supplied as well as all of the input data. Running the MIKE software requires a license. Read more about obtaining a license and about academic licensing. If you already have a license, read the step-by-step guide to learn how to run simulations.

# 3D model of Øresund

The coastal hydrodynamics of Øresund are primarily driven by the surge component introduced at the two open boundaries (one in North and one in South) with tidal ranges of 10-20 cm. The meteorological forcings such as air pressure and wind have only a minor effect. Modelling the hydrodynamics of this area is interesting for many reasons, e.g. surveillance of bacteria in bathing water and sediments around constructions and infrastructure.

## Model Setup

The Øresund mesh consists of triangular elements in a higher resolution close to the coast than on deeper waters, and for each node a depth value is assigned by interpolating the EMODnet bathymetry data. The mesh is shown in the figure below. It consists of a total of 3320 elements covering the area of approximately 2000 $km^2$.




### Forcings

The model is forced by meteorological inputs and boundary conditions supplied from a larger regional model. The meteorological inputs are:

- wind velocities and air pressure,
- air temperature and relative humidity,
- cloud coverage, precipitation and sea-ice cover.

The boundary conditions supply:

- vertical profile of water currents, temperature and salinity on the North and South boundary,
- water levels on the north and south boundary.

The outputs from the model are given as:

- a 3D timeseries of the current velocities (Eastward, Northward, and vertical directions), density, temperature and salinity.
- a 2D timeseries of the water surface elevations and depth-averaged current velocities.

The 2D timeseries contains values for each mesh element, whereas the 3D timeseries also covers the coordinates in the vertical direction.

## Validation

> **Template Note:** Describe the validation process for the model, including: - Observation Coverage - Validation approach - Skill metrics used - Comparison to operational model (if applicable)
>
> Validation scripts or notebooks should be located in: `code/model_validation_*.ipynb`

## Data Sources

> **Template Note:** Describe the edits that have been made to the original data. This could, for example, include outlier analysis and imputation. In the following table adapt the columns to fit the data sources you have used. Add rows as needed.

| Data | Source | Citation | License |
|---|---|---|---|
| e.g. Bathymetry | Provider + URL | How to cite the data | License type of license link |
| e.g. Forcing | Provider + URL | How to cite the data | License type of license link |